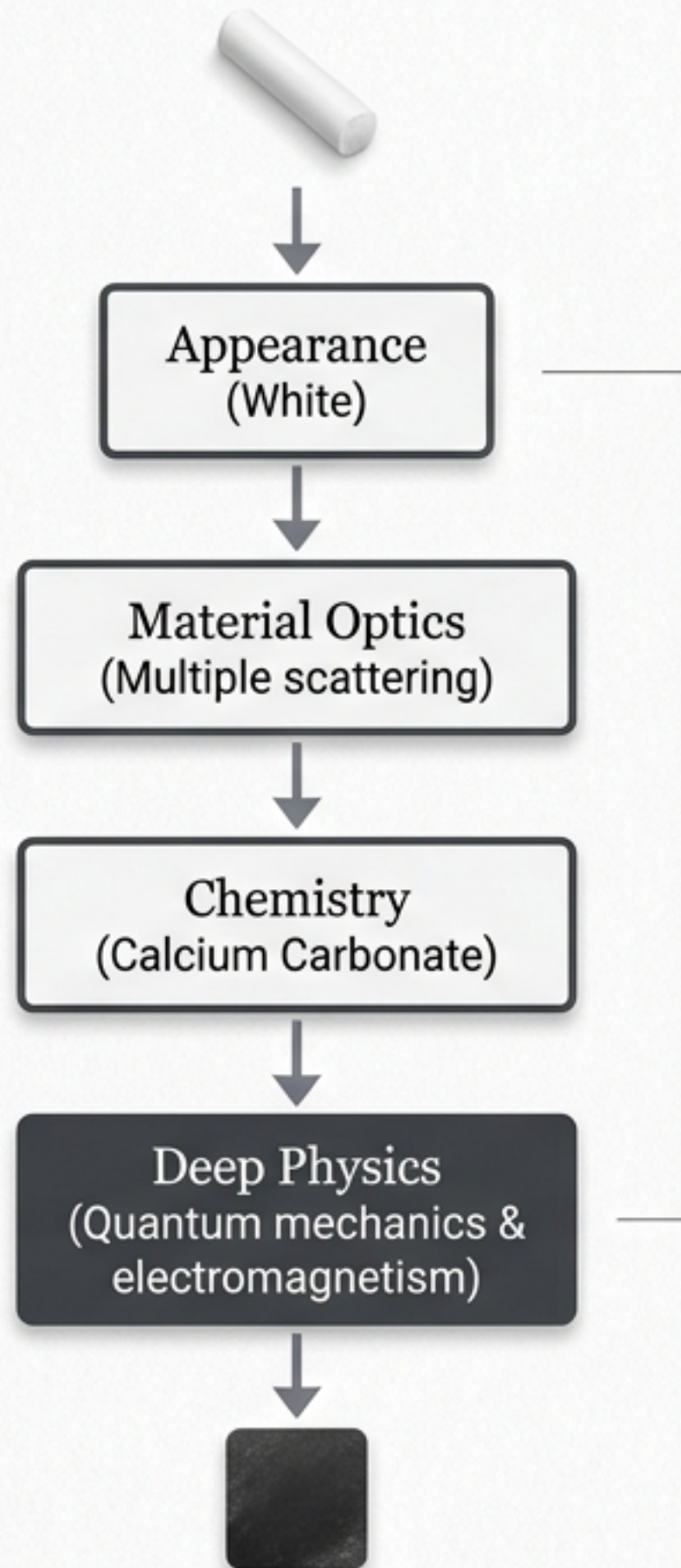




Why Are Leaves Green?

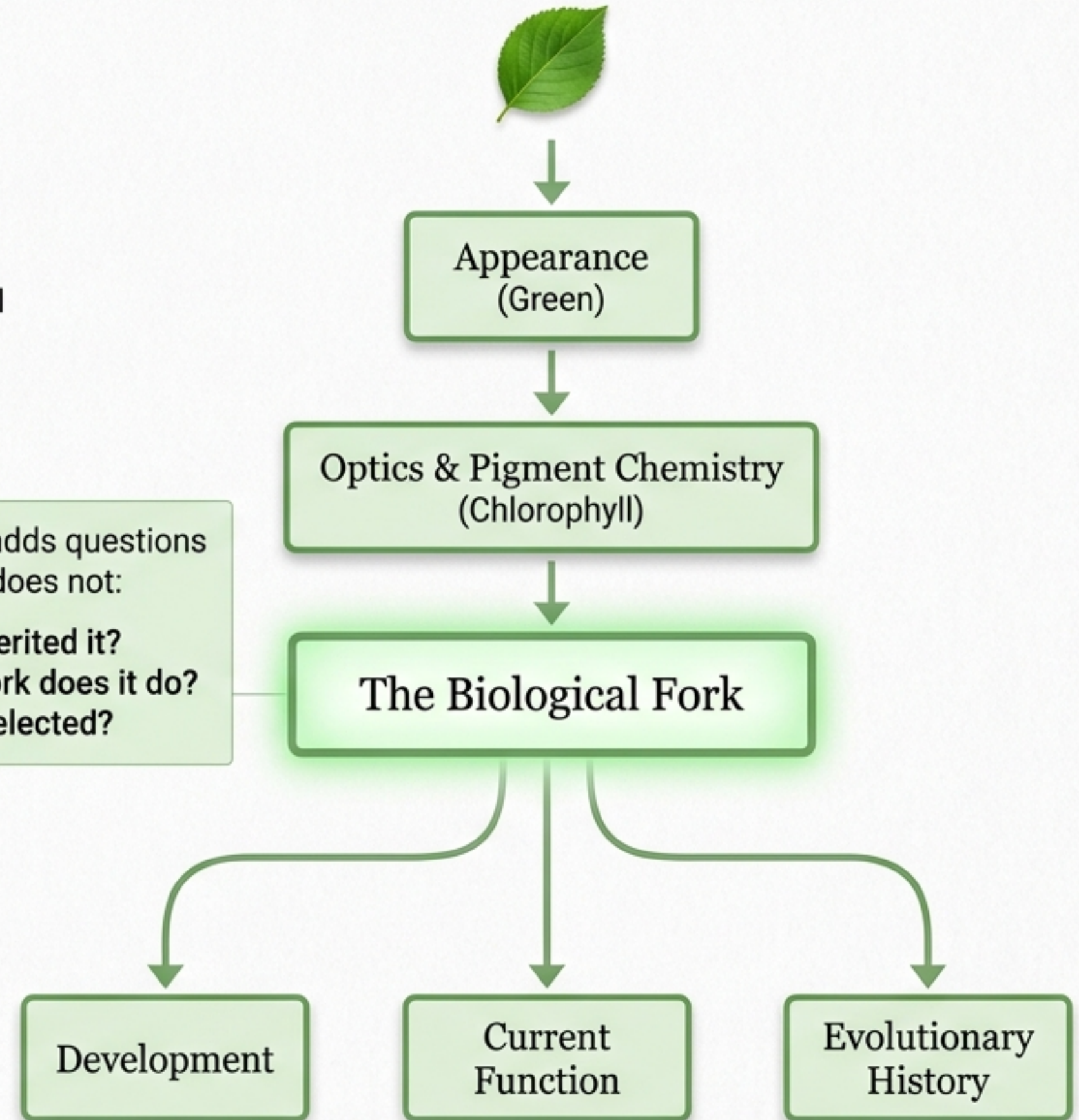
A Causal Descent. Where physical explanation ends, evolutionary storytelling begins.

The color is easy to explain. The history of the color is not. That gap is not a failure of science. It is an invitation to ask what kind of answer a piece of evidence can actually support.



Weinberg's descent connects everyday appearance to general physical principles.

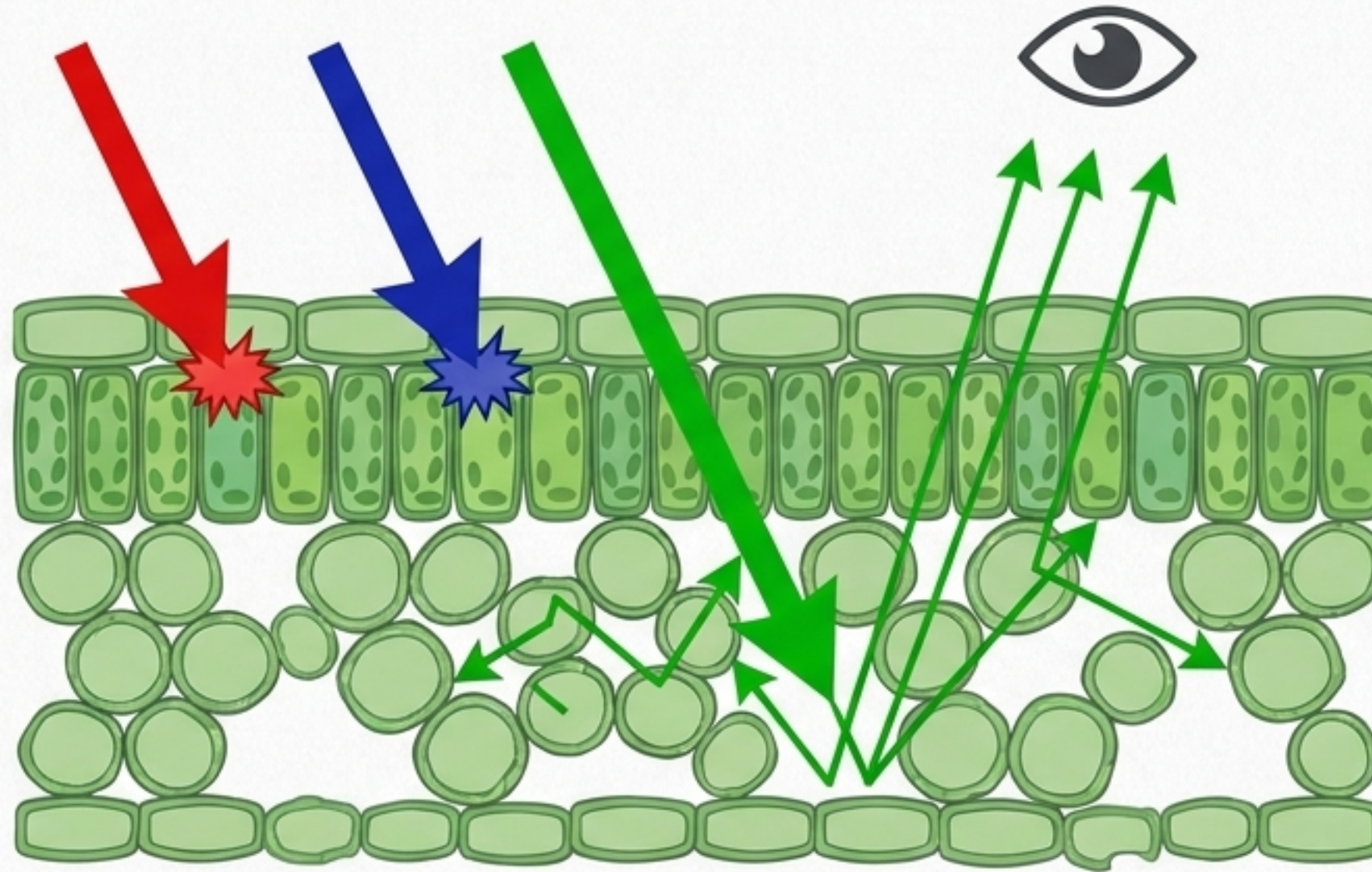
Biology adds questions physics does not:
Who inherited it?
What work does it do?
Was it selected?



Level 1 & 2: Optics and Pigments

[ESTABLISHED MECHANISM]

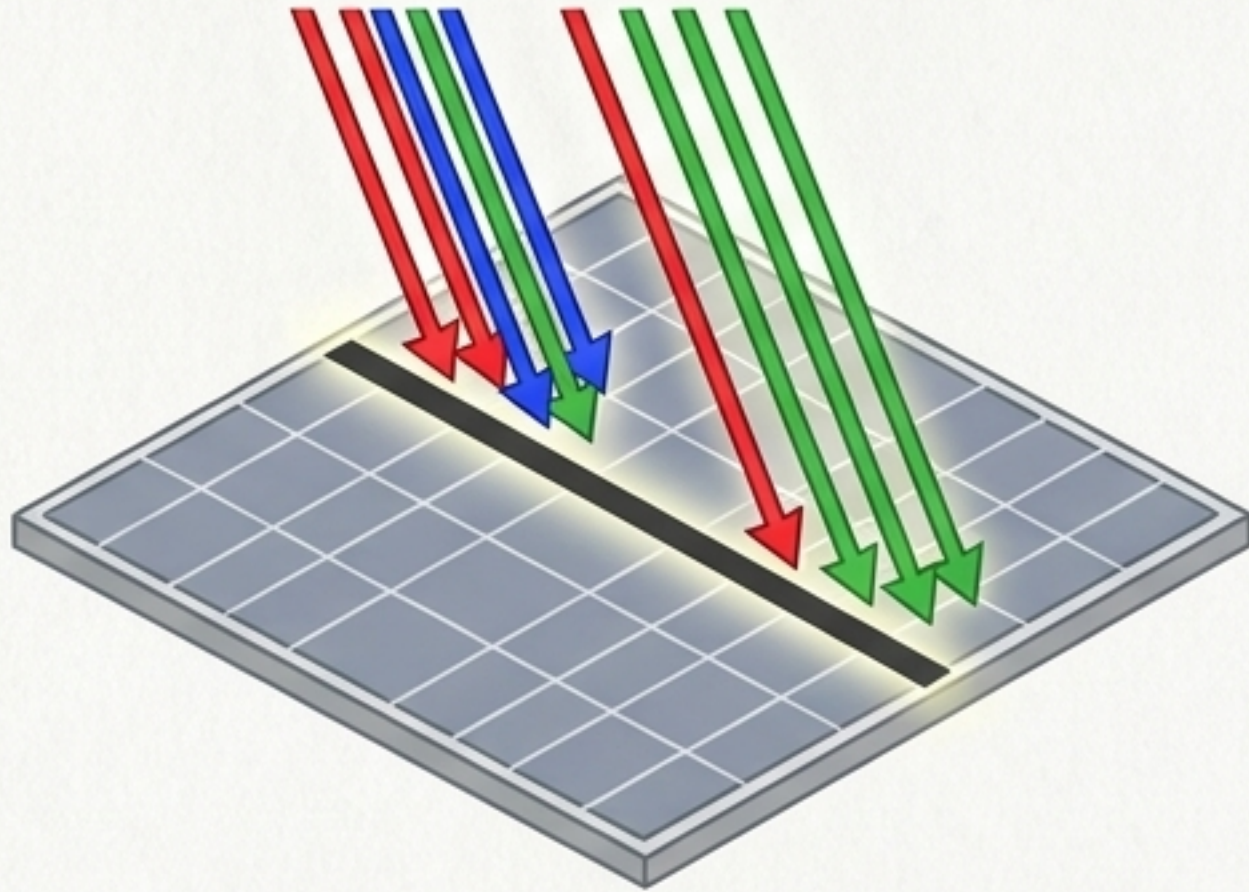
Leaves look green because their spectral absorption, scattering, and transmission return relatively more green light to an observer.



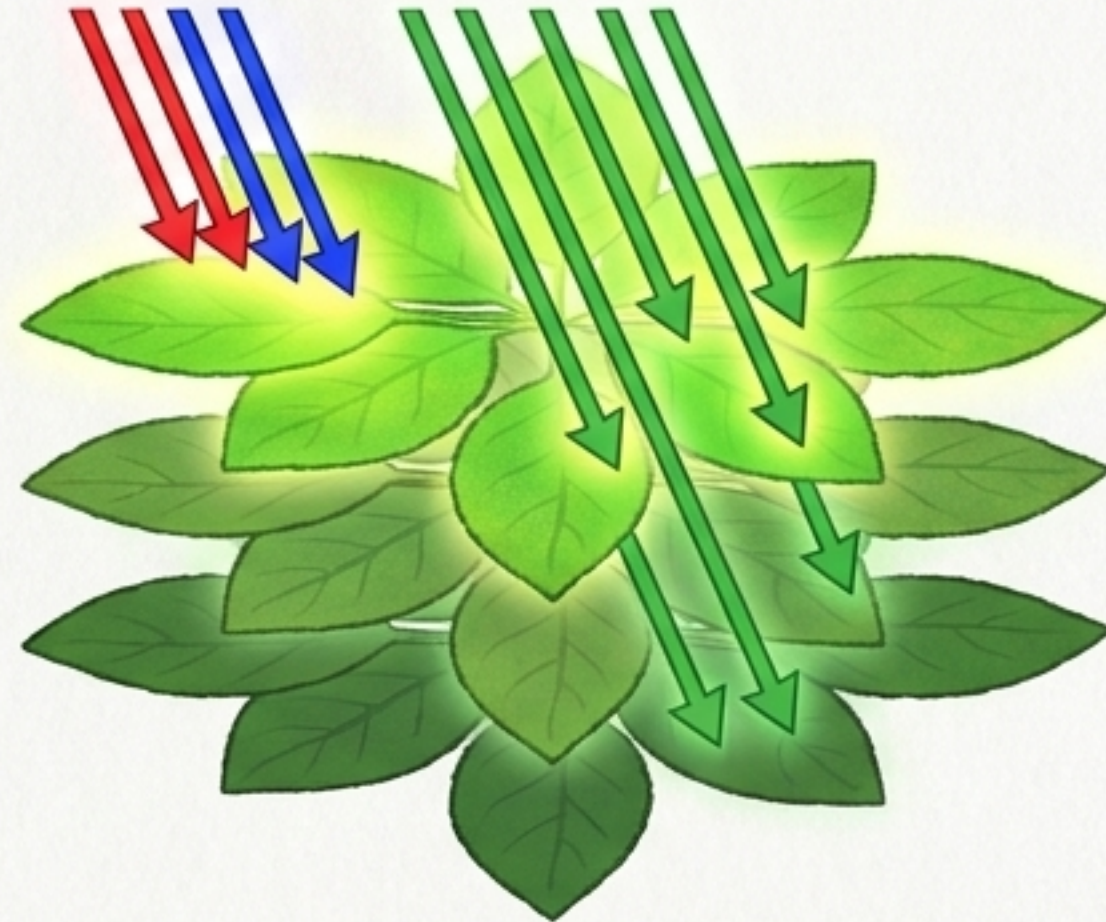
Plants do *not reject* green light. They simply absorb it less strongly than red and blue across the visible range.

Level 3: A Leaf is Not a Flat Solar Panel

Photosynthesis has **bottlenecks**. At noon, a sun leaf receives more photons than its biochemistry can handle. Weak surface absorption allows green photons to bypass saturated upper leaves and drive photosynthesis deeper in the canopy.



Maximum absorption by the first surface.



Maximum carbon gain for the whole plant.

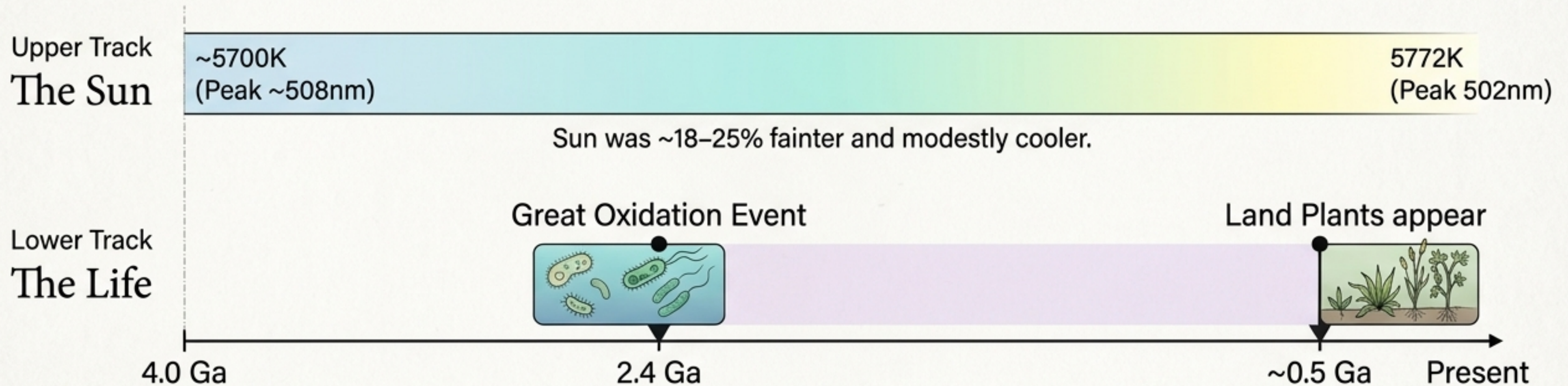
The Logical Leap: Useful Now $\neq$ Selected For That Use

MYTH	FACT
Plants reflect green light because they cannot use it.	False. Leaves absorb and use a great deal of green light, especially deeper in tissue where red and blue cannot reach.
Because avoiding excess light is useful now, plants originally evolved to reject green to prevent solar damage.	Unproven. A feature can arise as a side effect and become useful later. Present-day photoprotection does not automatically reconstruct the original selection pressure. (Gould & Lewontin, 1979)

Evolutionary biologists warn against turning every agreeable present effect into an origin story.

The Solar Spectrum Chronology Test

Wien's Law dictates the Sun's energy peak is in the green-cyan range (~502 nm).
Do leaves reject this peak?



The dates do not match a terrestrial tale. Oxygenic photosynthesis evolved underwater in microbes billions of years before land plants existed. Modern plants inherited an ancient system; they did not adapt from scratch to a naked solar spectrum on land.

Alternative Histories: Managing Unfinished Answers

Evidence Dossier Files [SPECULATIVE HYPOTHESIS]

Purple Earth

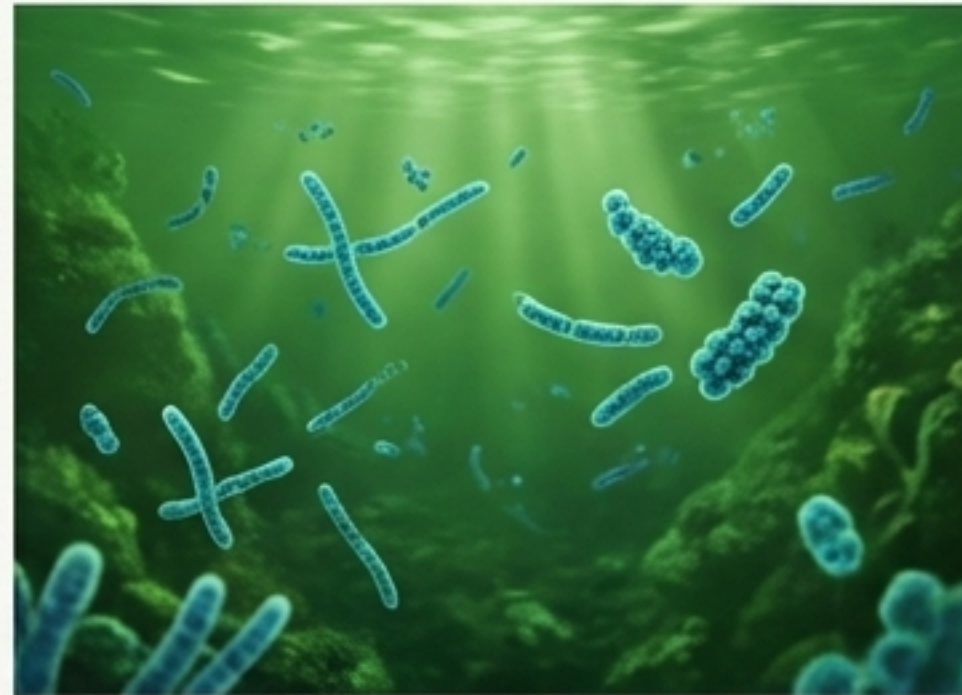


Retinal-based microbes dominated early Earth, forcing chlorophyll users into red/blue margins. (DasSarma & Schwieterman, 2021)

Biochemical possibility is not proof of global dominance.

Evidence Dossier Files [PLAUSIBLE MODEL]

Green Water

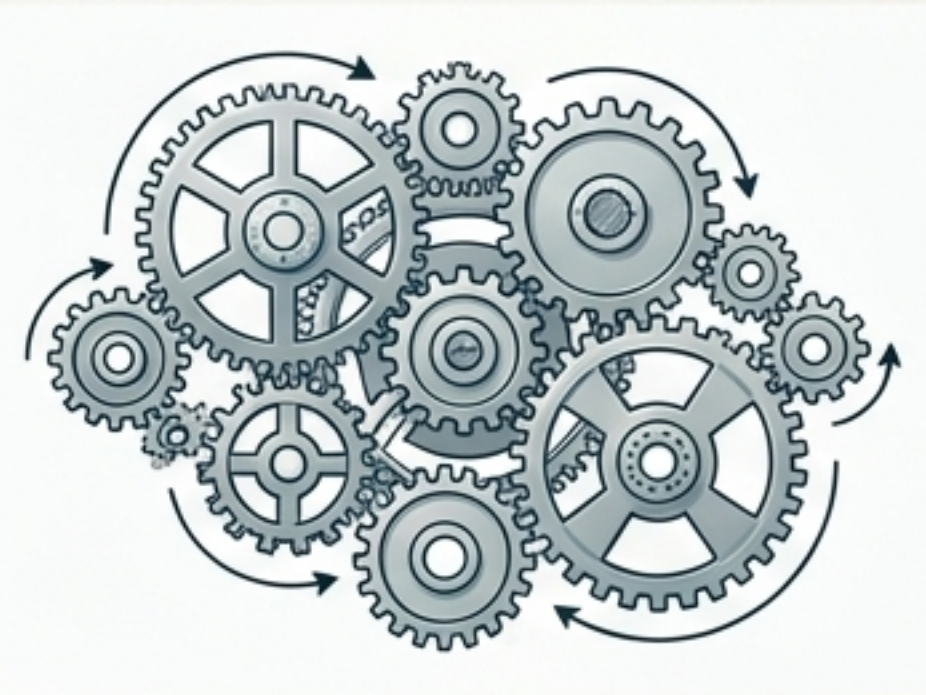


Archean underwater environments were iron-rich and green-lit, favoring cyanobacteria that adapted to use green light. (Matsuo et al., 2025)

Explains aquatic ancestors, not directly vascular plants.

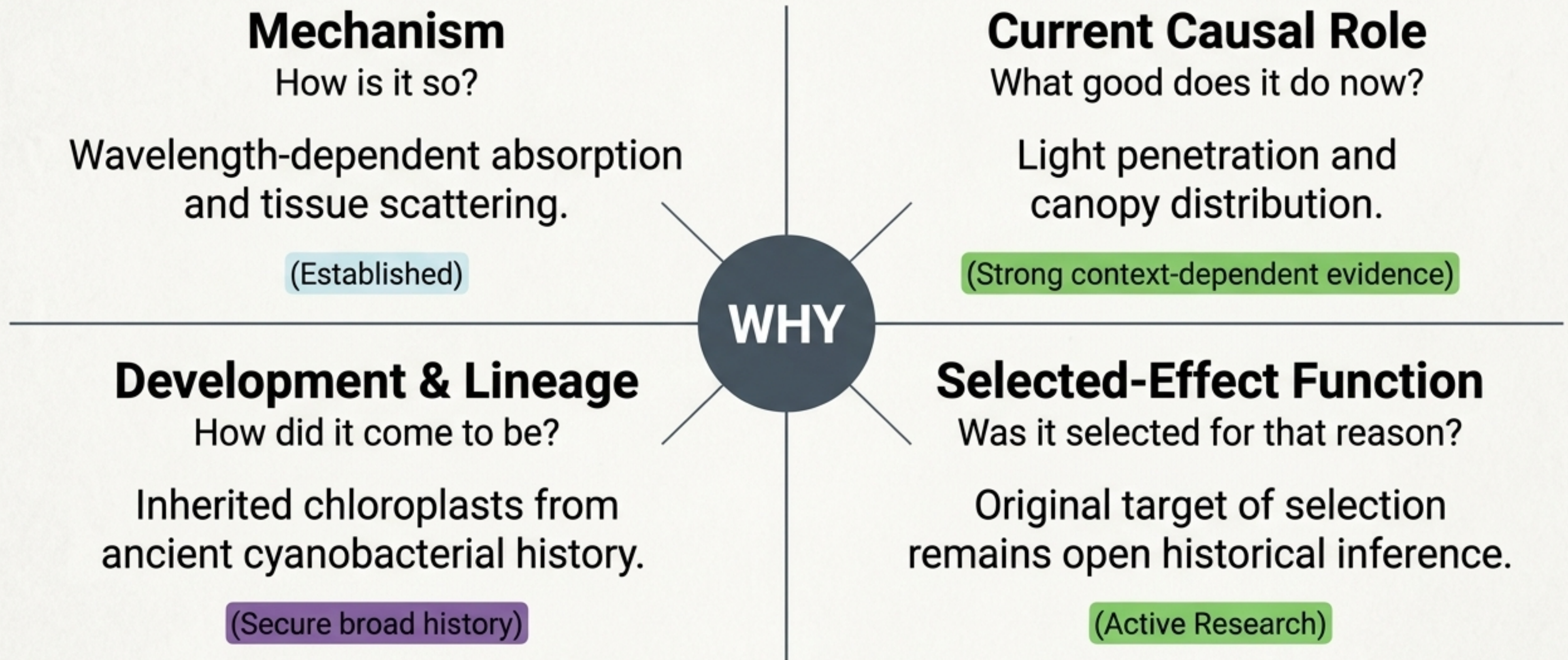
Evidence Dossier Files [VERY LIKELY]

Path Dependence



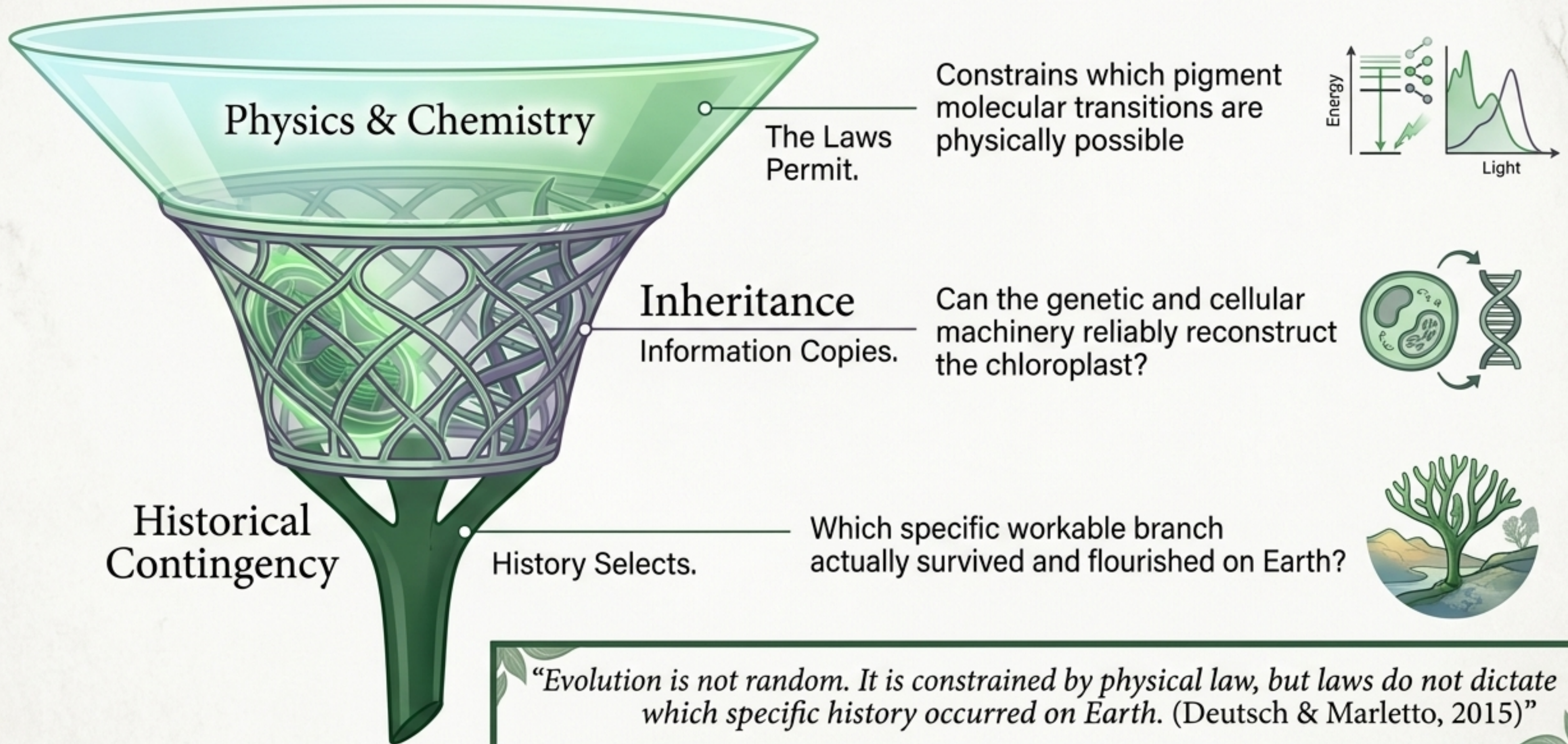
Once a complex photosystem coevolved, wholesale replacement was difficult. Selection optimized locally (noise-canceling/stability) rather than replacing the system. (Arp et al., 2020)

The Tinbergen Framework: Deconstructing the “Why”



The word *why* often smuggles four different scientific jobs into one sentence. We must separate them

Laws Permit; Histories Select



How Researchers Turn Stories Into Tests

1. Measure Mechanism.



Measure absorption vs. transmittance.

2. Change the System.



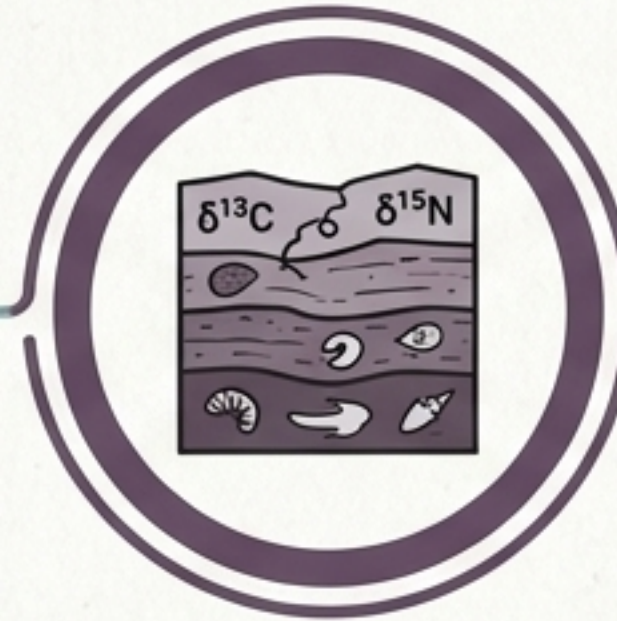
Create engineered mutants/transgenics to test causal effects.

3. Change the Scale.



Combine cuvette measurements with 3D radiative-transfer canopy models.

4. Reconstruct the Past.



Use phylogenies and geological proxies for ocean chemistry.

5. Make Models Risk Failure.



A good explanation must be difficult to vary. If you alter the canopy geometry in a model, it must predict exactly how the absorption profile will fail.

The Danger of the “Silent Upgrade”

Useful Compression

“Leaves look green because chlorophyll absorbs red/blue more strongly.”

Status: Serviceable mechanism.

The Silent Upgrade

“Plants do not use green light.”

Status: FALSE.

Story as Cause

“Plants evolved to reject green so sunlight wouldn’t damage them.”

Status: UNPROVEN. Plausible contribution promoted into an established origin story.

Warning Callout

School knowledge must be compressed. The danger begins when a useful compression is taught as if it answered a different question.

The Provenance Card (A Tool for Accountable Trust)

Claim: _____

(Smallest exact proposition)

Kind of Why: _____

(Mechanism, History, Current Role, Selection?)

Evidence: _____

(Measurement, Model, Fossil?)

Scope: _____

(Which species? Light intensity? Environment?)

Confidence: _____

(Established, Supported, Speculative?)

Source Chain: _____

(Where did this originate?)

The educational alternative to authoritarian certainty is not ‘believe whatever you like.’
It is accountable trust: matching the strength of a claim to the strength of its evidence.

The Composite 'Because'

“Why is the leaf green? Because photons scatter. Because inherited chloroplasts use green light deeper in the tissue. Because evolution built upon a workable ancestral system.”

“It does not do harm to the mystery to know a little about it.” — Richard Feynman

Knowledge is not weakened when we show its joints. A statement becomes more trustworthy when we can see which part is observed, which inferred, which modelled, and which is still waiting for a better test.